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Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 4, Semester 1 (2024)

IT4090 – Cloud Computing

Duration: 2 Hours

June 2024

Instructions to Candidates:

- This paper is preceded by 10 minutes reading period. The supervisor will indicate when answering may commence.
- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total mark for the paper is 100.
- This paper contains 4 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1 (25 marks)

a) Explain what "**Cloud Computing**" is and its key benefits. (4 marks)

b) What do you mean by a service model in Cloud Computing? Differentiate the three service models in Cloud Computing. (4 marks)

c) 'ABC' is a software company which handles various services. The company has a small data center at its main branch where the company's servers and data storage are located. All the staff of the company use different types of personal computers to connect with the company data center.

The directors of the company have received a report from a group of consultants that recommends the following.

- The company should consider using Office 365 as a SaaS service for office productivity applications and email.
- Company's some of the software could be run as a PaaS service for internal users but could also be supplied as a platform for other users to access as a service.
- The company can move some of its aging computing infrastructure to an IaaS service to avoid the capital cost of replacement.

i) Discuss the potential challenges that arise if you decide to continue with on-premises resources. (4 marks)

ii) Explain what you can do about the second suggestion given by the consultants regarding the software. (6 marks)

iii) Company management decides that the infrastructure should be created in a public cloud. Consider the cloud service provider as AWS or Azure and briefly explain what steps you would take to secure the infrastructure. (7 marks)

Question 2 (25 marks)

a) Explain the term "***Software Defined Security***". (3 marks)

b) IAM is a framework of policies and technologies to ensure users/process have the appropriate access to technology resources. Discuss three (3) IAM components. (5 marks)

c) Multi Factor Authentication (MFA) is a multi-step account login process that requires users to enter more information than just a password.

i. Describe three factors of multi factor authentication. (3 marks)

ii. Discuss the importance of multi factor authentication. (4 marks)

d) Differentiate Symmetric and Asymmetric encryption with aid of diagrams. (5 marks)

e) With the adoption of the cloud comes the need to ensure the organization's cloud security strategy that can protect against the top threats to cloud security. Do you agree with this? Justify your answer. (5 marks)

Question 3 (25 marks)

a) What is Microservices architecture? Briefly discuss in your own words. (4 marks)

b) State three (3) characteristics of Microservices and briefly explain each. (4 marks)

c) Differentiate Microservices architecture with Monolithic Architecture, with at least three (3) points. (6 marks)

d) Explain what **Function as a service (FaaS)** is and its key features. (4 marks)

e) Justify how the use of microservices, containers, and DevOps practices contributes to building and maintaining cloud-native architectures. Additionally, provide examples of tools or technologies commonly employed in the development and deployment of cloud-native applications. (7 marks)

Question 4 (25 marks)

a) Describe the three main barriers for IT Innovation. (3 marks)

b) Explain the fundamental reasons why organizations consider migrating their IT infrastructure to the cloud. (4 marks)

c) Explain the term "Cloud Well-Architected Framework" and evaluate how the key pillars of the Cloud Well-Architected Framework contribute to building robust cloud architectures. (5 marks)

d) Define the concept of Multi cloud integration and discuss advantages of multi cloud advantages. (3 marks)

e) "Cloud Computing plays a vital role in modern enterprises." Do you agree with this statement. Justify your answer. (5 marks)

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